Aloft (Grubbs)

Aloft is a granular insecticide used for turf and lawn applications. Apply directly to soil or turf according to label directions, using a spreader for even coverage. Water in after application to activate the product. Avoid applying to areas where runoff could reach waterways. Aloft primarily targets grubs and other soil-dwelling turf pests.

Safety Data Sheet - GHS

1. IDENTIFICATION: CHEMICAL PRODUCT AND COMPANY

PRODUCT NAME: Aloft® LC G Insecticide
EPA REGISTRATION NUMBER: 59639-215
VC NUMBER(S): 1992
PRODUCT DESCRIPTION: Insecticide
Aloft isaregisteredtrademarkofValent U.S.A. Corporation

M ANUF ACT URER/ DIST RI BUT OR
VALENTU.S.A.CORPORATION
P.O. Box 8025
1600 Riviera Avenue, Suite 200
Walnut Creek, CA 94596-8025

EMERGENCY TELEPHONE NUMBERS
HEALTHEMERGENCY OR SPILL(24hr):
(800) 892-0099
TRANSPORTATION (24 hr.): CHEMTREC
(800) 424-9300 or (202) 483-7616

PRODUCT INFORMATION
AGRICULTURALPRODUCTS:(800)682-5368

The current SDS is available through our website (www.valent.com), or by calling the product information numbers listed above.

2. HAZARDS IDENTIFICATION

This product is an EPA FIFRA registered pesticide. Some classifications on this SDS are not the same as the FIFRA-required classifications on the product label. Certain sections of this SDS are superseded by federal law under EPA FIFRA for a registered pesticide. Please see Section 15, REGULATORY INFORMATION for an explanation.

Classification - (per U.S. OSHA 29 CFR 1910.1200 (Hazcom 2012))

Not a dangerous substance or mixture according to the Globally Harmonized System (GHS).

Label elements

EMERGENCY OVERVIEW

WARNING

Precautionary statements

Prevention

None.

Response

Eyes IFIN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
Skin IFON SKIN: Gently wash with plenty of soap and water.
Inhalation IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
Ingestion IF SWALLOWED: Rinse mouth. DO NOT induce vomiting.
FIRE Incase of fire: Use CO2, dry chemical, or foam to extinguish.
Spill Collect spillage.

Storage
None.
Disposal
None.

Hazards not otherwise classified (HNOC)
Other Information
• Toxic toaquaticlife
• Very toxic to aquatic life with long lasting effects

99.625%ofthe mixtureconsists of ingredient(s) of unknown toxicity

For information on Transportation requirements see Section 14.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical Name	CAS Number	Weight/ Percent	TRADE SECRET
Clothianidin	2108 80-9 2-5	0.1 - 0.5	
Bifenthrin	82657-04-3	0.1 - 0.2	
Others	No CAS#	97 - 100	

Other ingredients, which may be maintained as trade secrets, are any substances other than an active ingredient contained in this product. Some of these may be hazardous, but their identities are withheld because they are considered trade secrets. The hazards associated with the other ingredients are addressed in this document. Specific information on other ingredients for the management of exposures, spills, or safety assessments can be obtained by a treating physician or nurse by calling **(800) 892-0099** at any time.

4. FIRST AID MEASURES

EMERGENCY NUMBER (800) 892-0099
Have the product container or label withyou when calling a poison control center or doctor, or going for treatment. You may also contact **1-800-892-0099** foremergency medical treatment information.

EYE CONTACT:
Hold eyeopen andrinse slowly and gently with water for 15-20 minutes. Remove contact lenses, if present, after the first 5 minutes, thencontinue rinsing eye. Call a poison control center or doctor for treatment advice.
SKIN CONTACT:
Take off contaminated clothing. Rinse skin immediately with plenty of water for 15-20 minutes. Call a poison control center or doctor fortreatment advice.

INGESTION:
Call apoisoncontrol center or doctor immediately for treatment advice. Have person sip a glass of water if able to swallow. DONOT induce vomiting unless told to do so by the poison control center or doctor. Do not give anything to an unconsciousperson.

INHALATION:
Moveperson tofresh air. If person is not breathing, call 911 or an ambulance, then give artificial respiration, preferably by mouth-to-mouth, if possible. Call a poison control center or doctor for further treatment advice.

NOTES TO PHYSICIAN:
None

5. FIRE FIGHTING MEASURES

Flash point °C Notapplicable
EXTINGUISHING MEDIA: Smallfires: Foam, carbon dioxide, or dry chemical. Large fires: Water Fog
FLAMMABLE LIMITS IN AIR - LOWER (%): Not applicable
FLAMMABLE LIMITS IN AIR - UPPER (%): Not applicable

NFPA RATING:

Health:	2 1 0
Flammability:	None
Reactivity:	
Special:	

(Least-0, Slight-1, Moderate-2, High-3, Extreme-4). These values are obtained using professional judgement. Values were not available in the guidelines or published evaluations prepared by the National Fire Protection Association, NFPA.

FIRE FIGHTING INSTRUCTIONS: Products of combustion from fires involving this material may be toxic. Avoid breathing smoke and mists. Avoid personnel and equipment contact with fallout and runoff. Minimize the amount of water used for fire fighting. Do not enter any enclosed area without full protective equipment, including self-contained breathing equipment. Contain and isolate runoff and debris for proper disposal. Decontaminate personal protective equipment and fire fighting equipment before reuse.

6. ACCIDENTAL RELEASE MEASURES

VALENT EMERGENCY PHONE NUMBER: (800) 892-0099
CHEMTREC EMERGENCY PHONE NUMBER: (800) 424-9300
OBSERVE PRECAUTIONS IN SECTION 8: PERSONAL PROTECTION

Stop the source of the spill ifsafeto do so.Contain thespillto prevent furthercontaminationofthe soil, surface water, or ground water. For additional spill response information refer to the North American Emergency Response Guidebook.

UN/NA NUMBER: Notapplicable **EMERGENCY RESPONSE GUIDEBOOK NO.:** Not applicable

FOR SPILLS AND LEAKS

CONTAINMENT:
Small spills: Sweep up carefully while avoiding the formation of a dust cloud. Place in an approved chemical waste containerfordisposal. Rinse spill area with a small amount of soapy water. Contain and absorb the rinsate with inert absorbents and place into the same disposal container. The area can be washed with water to remove the last trace residue. Do not allow spilled material to enter any body of water.

CLEANUP: Clean up spill immediately. Vacuum or sweep up material and place in a chemical waste container. Wash area with soap and water. Pick up wash liquid with additional absorbent and place in a chemical waste container.

7. HANDLING AND STORAGE

END USER MUST READ AND OBSERVE ALL PRECAUTIONS ON PRODUCT LABEL.

H AN DLING:

Use onlyinwell ventilated area. Avoidcontact with skin andeyes. Do not breathe spray mist. Keep pesticide in originalcontainer. Wash thoroughly with soap and water after handling and before eating, drinking, chewing gum, using tobacco or using the toilet. Remove and wash contaminated clothing before reuse.

STORAGE:

Storeinwell ventilated area. Keep container tightly closed. Keep pesticide in original container. Do not store or transportnear food or feed. Do not contaminate food or feed. Do not put concentrate into food or drink containers. Do notdilute concentrate in food or drink containers. Store in a cool, dry place, out of direct sunlight.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

END USER MUST READ AND OBSERVE ALL PRECAUTIONS ON PRODUCT LABEL.

EYES & FACE: Donotgetthismaterialin youreyes.Eyecontactcanbe avoidedbywearing protective eyewear.

RESPIRATORY PROTECTION: Usethis material onlyinwell ventilated areas.Ifoperating conditions result in airborneconcentrationsofthismaterial, the useof an approved respirator is recommended.

SKIN & HAND PROTECTION: Do not get on skin or clothing. Skin contact should be minimized by wearing protective clothing including coveralls worn over short-sleeved shirt and short pants, socks, chemical-resistant footwear and chemical-resistant gloves. Remove contaminated clothing.

ENGINEERING CONTROLS: Use in a well ventilated area.

EXPOSURELIMITS

Chemical Name	ACGIH Exposure Limits	OSHA Exposure Limits	Manufacturer's Exposure Limits
Clothianidin	None	None	None
Bifenthrin	None	None	None
Others	None	None	None
			None

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	Solid		
Appearance	Granules	**Odor**	Odorless
Color	Gray	**Odor threshold**	No information available

PROPERTIES	**Values**	Remarks •Method
pH	7 - 8 No information available No information available	
Melting point/freezing point	Not applicable	
Boiling point/boiling range	No information available	
Flash point	No information available	
Evaporation rate		
Flammability (solid, gas)		

Flammability Limits in Air	
Upper flammability limits	Not applicable
Lower flammability limit	Not applicable
Vapor pressure	No information available
Vapor density	No information available
Specific Gravity	No information available
Water solubility	Negligible
Solubility in other solvents	No information available
Partition coefficient	No information available
Autoignition temperature	No information available
Decomposition temperature	No information available
Viscosity	No information available
Explosive properties	No information available
Oxidizing properties	No information available
Density	0.85 g/mL
Bulk density	No information available

10. STABILITY AND REACTIVITY

Reactivity

Notreactive

Chemical stability

Stableunderrecommended storageconditions.

Possibility of Hazardous Reactions

Noneundernormalprocessing.

Conditions to avoid

Extremesoftemperatureand direct sunlight.

Incompatible materials

Noneknownbasedoninformation supplied.

Hazardous Decomposition Products

Noneknownbasedoninformationsupplied.

11. TOXICOLOGICAL INFORMATION

ACUTE TOXICITY:

Oral Toxicity LD 50 (rats)	>5000	mg/kg	EPA Tox Category	III
Dermal Toxicity LD 50 (rabbits)	>5000	mg/kg	EPA Tox Category	IV
Inhalation Toxicity LC 50 (rats)	>2.05	mg/L	EPA Tox Category	IV
Eye Irritation (rabbits)	Slightly irritating		EPA Tox Category	IV
Skin Irritation (rabbits)	Slightly irritating		EPA Tox Category	IV
Skin Sensitization (guinea pigs)	Non-sensitizer		EPA Tox Category	Not applicable

CARCINOGEN CLASSIFICATION

Chemical Name	IARC	OSHA - Select Carcinogens	NTP Carcinogen List
Clothianidin	Not listed	Not Listed	Not listed
Bifenthrin	Not listed	Carcinogen	Not listed
Others	Not listed	Not listed	Not listed

For a summary of the potential for adverse health effects from exposure to this product, refer to Section 2. For information regarding regulations pertaining to this product, refer to Section 15.

12. ECOLOGICAL INFORMATION

The following data were developed with: Clothianidin Technical

AQUATIC ORGANISM TOXICITY: Clothianidin Technical is very toxic to aquatic life with long lasting effects:

96-hour LC 50 Rainbow Trout >100 mg/L
96-hour LC 50 Bluegill >100 mg/L
96-hour LC 50 Sheepshead minnow >100 mg/L
48-hour LC 50 Water flea = 40 mg/L
96-hour LC50 Mysid = 0.053 mg/L

OTHER NON-TARGET ORGANISM TOXICITY: This product is toxic to bees exposed to direct treatment, and for more than 5 days after, on blooming crops or weeds. Do not apply this product while bees are actively visiting the treatment area.

OTHERENVIRONMENTALINFORMATION:
May cause long-term adverse effects in the environment.

13. DISPOSAL CONSIDERATIONS

END USERS MUST DISPOSE OF ANY UNUSED PRODUCT AS PER THE LABEL RECOMMENDATIONS.

PRODUCTDISPOSAL: Wastes resulting from the use of this product may be disposed of on site or at an approved waste disposal facility.
CONTAINER DISPOSAL: Non-refillable container. Do not reuse or refill this container. Completely empty bag into application equipment. Then dispose of bag in a sanitary landfill.

DISPOSAL METHODS: Check government regulations and local authorities for approved disposal of this material. Dispose of in accordance with applicable laws and regulations.

14. TRANSPORTATION INFORMATION

DOT (ground) SHIPPING NAME: Not regulated for domestic ground transport by U.S. DOT

EMERGENCY RESPONSE GUIDEBOOK NO.: Not applicable

ICAO/IATA SHIPPING NAME: UN 3082 Environmentally Hazardous Substance, Solid, N.O.S. (Clothianidin, Bifenthrin), 9, III, Marine Pollutant

REMARKS: Single or inner packaging less than 5 L (liquid) or 5 Kg net (solids) excepted from Dangerous Goods regulations -- see UN Special Provision 375.

IMDG SHIPPING NAME: UN 3082 Environmentally Hazardous Substance, Solid, N.O.S. (Clothianidin, Bifenthrin), 9, III, Marine Pollutant

REMARKS: •Single or inner packaging less than 5 L (liquid) or 5 Kg net (solids) excepted from Dangerous Goods regulations – see IMDG 2.10.2.7
•For US shipping, Emergency Response Guidebook No. 171

EMS NO.: F-A,S-F

15. REGULATORY INFORMATION

EPA-FIFRA LABEL INFORMATION THAT DIFFERS FROM OSHA-GHS REQUIREMENTS:

This material is a pesticide product registered by the EPA under FIFRA and is subject to certain labeling requirements under federal pesticide law. These requirements may differ from the classification criteria and hazard information required by OSHA GHS for safety data sheets, and for workplace labels of non-pesticide chemicals. The following is the hazard information as required on the FIFRA pesticide label:

EPA FIFRA SIGNAL WORD: CAUTION

- *Harmful if swallowed, inhaled or absorbed through skin.*
- *Causes moderate eye irritation*
- *Avoid contact with eyes, skin and clothing.*
- *Avoid breathing vapors or spray.*
- *Keep out of reach of children.*

PESTICIDE REGULATIONS: All pesticides are governed under FIFRA (Federal Insecticide, Fungicide, and Rodenticide Act). Therefore, the regulations presented below are pertinent only when handled outside of the normal use and applications of pesticides. This includes waste streams resulting from manufacturing/formulation facilities, spills or misuse of products, and storage of large quantities of products containing hazardous or extremely hazardous substances.
U.S. FEDERAL REGULATIONS: Ingredients in this product are reviewed against an inclusive list of federal regulations. Therefore, the user should consult appropriate authorities. The federal regulations reviewed include: Clean Water Act, SARA, CERCLA, RCRA, DOT, TSCA and OSHA. If no components or information is listed in the space below this paragraph, then none of the regulations reviewed are applicable.

Bifenthrin
SARA313 Chemicals 1.0% de minimis concentration

SARA (311, 312):
ImmediateHealth: Yes
Chronic Health: Yes
Fire: Yes
Sudden Pressure: No
Reactivity: No

STATE REGULATIONS: Each state may promulgate standards more stringent than the federal government. This section cannot encompass an inclusive list of all state regulations. Therefore, the user should consult state or local authorities. The state regulations reviewed include: California Proposition 65, California Directors List of Hazardous Substances, Massachusetts Right to Know, Michigan Critical Materials List, New Jersey Right to Know, Pennsylvania Right to Know, Rhode Island Right to Know and the Minnesota Hazardous Substance list. For Washington State Right to Know, see Section 8 for Exposure Limit information. For Louisiana Right to Know refer to SARA information listed under U.S. Regulations above. If no components or information is listed in the space below this paragraph, then none of the regulations reviewed are applicable.

Bifenthrin
NJRightTo Know 3194

For information regarding potential adverse health effects from exposure to this product, refer to Sections 2 and 11.

16. OTHER INFORMATION

REASON FOR ISSUE: Create a new MSDS for an existing material. Added the EPA registration number.
SDS NO.: 0482
EPA REGISTRATION NUMBER: 59639-215
REVISION NUMBER: 1
REVISION DATE: 11/24/2015
SUPERCEDES DATE: None
RESPONSIBLE PERSON(S): Valent U.S.A. Corporation, Corporate EH&S, (925) 256-2803

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This Safety Data Sheet (SDS) serves different purposes than and DOES NOT REPLACE OR MODIFY THE EPA-APPROVED PRODUCT LABEL (attached to and accompanying the product container). This SDS provides important health, safety, and environmental information for employers, employees, emergency responders and others handling large quantities of the product in activities generally other than product use as required by the Occupational Health and Safety Act (29 CFR 1910.1200, "Hazcom").

The product label provides information specifically for product use in the ordinary course. Use, storage and disposal of pesticide products is regulated by the EPA under the authority of FIFRA through the proudct label. All necessary hazard classification and appropriate precautionary use, storage, and disposal information is set forth on that label or labeling accompanying the pesticide or to which reference is made on the label. It is a violation of federal law to use an EPA-registered pesticide product in any manner inconsistent with its labeling.

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